

Software Engineer · AI Systems

Building AI technologies with AI tools.

I learned Computer Science first, then learned the inside of LLMs by hand — the substrate underneath and the tools on top. When an AI tool is wrong, I notice. When a model fails silently, I know how to diagnose it.

[See everything I've built >](#)

FOUNDATIONS

Hand-implemented transformer and GPT internals from scratch — **attention, backpropagation, tokenization, embeddings** — following Karpathy's **Neural Networks: Zero to Hero** end-to-end, before adopting AI-assisted development. Self-directed coursework in the math and ML underneath: **Stanford CS229**, MIT OpenCourseWare Linear Algebra, Differential Equations, Calculus, and MIT 6.034 Artificial Intelligence.

SELECTED PROJECTS

Manufacturing Reality Orchestrator 2026 PyTorch · Transformers · SHAP · Three.js Advisory layer reading four planning systems (PLM, ERP, MES, procedures) together to catch cross-system disturbances that stay inside every individual limit. Single transformer produces a slip prediction plus a ranked hypothesis queue via input attribution. Design stage. Demo > · Case Study > · How It Works >	Rocket Engine Health Orchestrator 2026 PyTorch · Transformers · Three.js Per-channel encoders fused by attention to catch coupled, sub-threshold faults that per-channel redlines are structurally blind to. Advisory above the deterministic redline, which keeps authority. Architecture, physics based data generator designed end to end. Design done, dev started; no model trained. Demo > · Case Study > · How It Works >	Turbofan Predictive Maintenance 2025–26 PyTorch · Transformers · Flask Dual-head Transformer for jet-engine anomaly detection and Remaining Useful Life. Diagnosed a total model collapse to a data signal-to-noise problem, not the model — rebuilt the data, then derived a cost-weighted operating point cutting missed failures from 586 to 13 on the held-out test set. See the Case Study >
AskMickey — AI Disney Assistant 2025 Flutter · Dart · Gemini · 4 platforms Designed a classify-then-route dispatcher on Gemini — MoE-inspired but deterministic — routing each query to one of eight handlers before any generative call, with safety paths resolved at classification time rather than left to the model. Shipped and live on four platforms from a single codebase. See the Case Study >	Weather: Physics vs. ML 2026 JS · Python · BigQuery · ECMWF AIFS A live evaluation system comparing GFS, ECMWF AIFS, and DeepMind WeatherNext, scored against Iowa Mesonet ASOS ground truth. The real discipline was the verification loop — caught two silent data-source bugs that would have corrupted the scoreboard. See the Case Study >	Pong from Pixels 2026 PyTorch · Gymnasium · CUDA Deep Q-network learning Pong from raw pixels on a 6 GB GPU. Network, replay buffer, exploration schedule, and training loop hand-written; emulator and autograd imported and named as such. Ongoing four-hour Saturdays, pre-registered floor and aspiration goals. Training not yet started. See the Case Study >

TECHNICAL SKILLS

AI / ML — PyTorch · Transformers · LLMs / RAG · Agentic AI · Multi-Task Learning · scikit-learn · **Languages** — Python · C · C++ · Java · JavaScript · Dart · Swift · SQL · **Frameworks** — Flask · Flutter · Pandas · NumPy · Matplotlib · **Infrastructure** — Linux / UNIX · Amazon EC2 · Apache · MySQL · **Tools** — Git · VS Code · Arduino · Raspberry Pi · **Practice** — Feature branches · PRs carrying reasoning · Versioned releases

EXPERIENCE — EPCOM CORPORATION

Machine Learning Software Engineer Intern · Summers 2024–2025
Built a Transformer-based predictive-maintenance system for multivariate time-series sensor data. Integrated retrieval-augmented LLMs via prompt engineering and structured outputs; built data pipelines and evaluation harnesses for iterative development and benchmarking.

Software Development Intern · Summers 2020–2023
Full-stack components of an Applicant Tracking System (Linux · Apache · MySQL · PHP · JavaScript · Bootstrap) on Amazon EC2; automated cron jobs with pattern recognition to ingest applicants from IMAP email.

EDUCATION

University of Florida · August 2025
B.S. Computer Science

St. Petersburg College
A.A., Cum Laude

Volunteer Coach · Town of Belleair
Basketball (Coach of the Year) & Tennis, ages 4–16